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Automatic calibration of agent-based public transit assignment path choice to count data ☆

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ABSTRACT

This work describes the calibration of a schedule-based transit assignment inside an iterative microscopic agent-based simulation. The calibration challenge implies that the behavioral rules should be modified in order to move the simulation closer to observed passenger counts. First, route choice set of agents is enriched with travel parameter utilities randomization. Secondly, the calibration interacts directly into the performance evaluation of individual daily plan of activities, so that the plan is also evaluated for its contribution to the count reproduction. In this way, appropriate plans from the calibration perspective can persist along simulation iterations. The Berlin public transport system with day-based counts is used as test scenario. The results show that the calibration approach can work with large scale scenarios, and that it is able to deal with the inter-temporal aspects implied by counts.

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1. Introduction

The implementation of a microscopic agent-based model for public transport simulation represents a helpful instrument for mobility research. If the simulation includes also a travel behavioral approach, it constitutes an advantageous analysis tool for the knowledge of passengers' route decisions and to recognize the effects of those decisions on the whole transport system.

In microscopic models, a special challenge has been calibration. More technically: Given a set of macroscopic observations, how should the physical or behavioral microscopic rules of the agent-based simulation be modified in order to move the simulation closer to the observations? The example considered here is a situation where one has daily passenger counts for certain or all transit lines, demand for 1% of the population from trip diaries, but no route information from the survey. The task is to find passenger connections and possibly modify the passenger demand such that the simulation matches the counts.

This topic belongs to a class of problems that are quite common in agent-based simulations. Agent-based simulations were always built around the notion of "emergence", that is, they are expected to be particularly useful where certain macroscopic properties, in our case congestion, vehicle overloading, and resulting delay patterns, cannot be derived in analytical ways from the microscopic input data (including the behavioral rules), and in consequence one needs to run the simulation

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in order to obtain them. However, because the connection from input data to emergent properties is by simulation, the mathematical connection is not as well established as in normal numerical modeling.

2. Related works

The use of smart cards, sensors or surveys as data sources for passenger measurements allows the availability of transportation usage information for research purposes (Pelletier et al., 2011; Min et al., 2016). Several works for demand estimation of large scenarios based on passenger counts can be found. Rongviriyapanich et al. (2000) use on–off count data for two bus routes in Tokyo to evaluate OD estimation techniques used originally for road networks. The Entropy Maximization Method is reported as the most practical of all considered techniques, if a prior OD matrix is available. In the same way, Fung (2005) uses boarding and alighting counts of the Hong Kong metro network for station-to-station OD matrix assignment calibration and validation. Random choice function coefficients generation methods in Monte Carlo simulation are also presented. Farrol and Livshits (1998) calibrate a scenario of the Greater Toronto Area based on a survey in 1996. Their results present the adjusted weights to access time, wait time, and the transfer penalty in an EMME/2 implementation. Parveen et al. (2007) present the G-EMME/2 tool to estimate also the Toronto transit network with volumes obtained from on-board surveys. A genetic algorithm represents chromosomes as sets of parameter values generated randomly. The minimization of a "misfit" function optimizes the travel time of the five parameters: boarding time, wait-time factor, wait-time weight, auxiliary time weight, and boarding-time weight. Lu (2008) uses automatically-obtained passenger boarding and alighting counts for a bus line in Columbus, Ohio to review five OD flow estimation methods. The results show that the output quality depends to a great extent on the quality of the base OD matrix. Li (2009) presents the statistical inference of large transit route matrices using on–off counts. Considering a given occupancy of a passenger on a stop, the probability of alighting on a posterior stop of the transit line is calculated with a Markov chain model. A Bayesian analysis draws inference about unknown parameters. Wahba and Shalaby (2011) calibrate the transit scenario of Toronto with the integration of a genetic algorithm engine. Passengers travel experiences are assessed. Thus, the learning mechanism adjusts waiting and in-vehicle time. Moreover, the calibration defines nine parameters related to trip purpose and vehicle type. Ji et al. (2015) present a heuristic expectation maximization model to estimate passenger OD flows using information from onboard surveys. The model is evaluated with empirical comparison to ground-truth observations and with the estimation realized with the iterative proportional fitting estimation procedure. The initial estimation is improved with the inclusion of larger boarding and alighting data vectors obtained with automatic passenger count technologies.

Most demand estimation models focus on aggregated OD flows. That is a disadvantage, considering the research opportunities that can be given with passenger mobility data on an individual level. The present work makes a contribution with a microscopic agent-based approach that addresses the estimation at a fully disaggregated level, catapulting the possibilities of detailed or collective travel behavior analysis.

3. Simulation methodology

An important challenge for the agent-based approach for the recent past has been computational: Finding an implementation that is both close to the agent-based concepts and fast enough for real world scenarios. For transit micro-simulation, the open source framework MATSim (Multi-Agent Transport Simulation) is employed in this work. Its capacity to simulate large scale scenarios and its modular architecture make it appropriate for calibration research. Its simulation approach follows these steps:

- Initialization: Synthetic travelers with activity-based day plans are assumed to be given; see below how they were generated for the present study. This includes generating suitable connections through the public transit network for each trip that is part of the passenger's plan. The calculation of routes and more transit simulation details are described by Rieser (2010). The routing process looks for least compound cost paths with a trade-off between walk time, waiting time, in-vehicle travel time, travel distance, and vehicle change counts. These travel preferences are represented respectively in an economic approach by the parameters presented in Table 1.
- The "effective value" includes the marginal Opportunity Cost of Time, which needs to be added to all "direct" marginal (dis)utilities of time (de Serpa, 1971; Jara-Diaz, 2003; Börjesson and Eliasson, 2014). The opportunity cost of time represents the implicit punishment for not performing an activity. In MATSim, it is approximately given by the parameter called "*performing*", and its default is −6/h. The utilities are taken as dimensionless quantities; −6/h, say, means "minus six utils per hour" or "minus 6/3600 utils per second". Note that the original approach weights all temporal contributions equally.
- Synthetic reality (also known as network loading): All plans are executed by moving agents in the multimodal network simultaneously in a simulation of the physical system ("synthetic reality") which includes a detailed simulation of the public transit system. The events triggered by the simulation are useful for modeling travel incidents such as boarding delays, failures because of overloading, and delays because of late incoming vehicles.

Table 1
Transit travel parameters.

Name for present paper	Name in MATSim config	Current MATSim default value	Effective value (including Opportunity Cost of Time)
Marginal Utility of Travel Time Walk (MUTTW)	travelingWalk	−6/h	−12/h
Marginal Utility of Transit Waiting Time (MUTWT)	waitingPt	−6/h	−12/h
Marginal Utility of Travel Time Transit (MUTTT)	travelingPt	−6/h	−12/h
Marginal Utility of Travel Distance Transit (MUTDT)	monetaryDistanceCostRatePt ∗ marginalUtilityOfMoney	0	0
Utility of Line Switch (ULS)	utilityOfLineSwitch	−1	−1

- Scoring: Each plan is scored based on its own performance in the synthetic reality. Following a utility-based approach, the utility $V(i)$ of a plan i is calculated as the sum of positive utilities (in a logarithmic form) achieved by carrying out activities, plus the sum of negative utilities of traveling between activities locations

$$V(i) = \sum_{act \in i} \beta_{perf} \cdot t^*_{act} \cdot \ln\, t_{perf,act} + \sum_{leg \in i} V_{tr,leg} \tag{1}$$

where
β_{perf} is the activity marginal utility at its typical duration.
t^*_{act} is the activity typical duration.
$t_{perf,act}$ is the activity duration in the simulation (in sec).
$V_{tr,leg}$ is the utility (typically negative) of a leg.
- Choice set modification: The synthetic travelers try to construct new plans given their experience from the previous simulated day (iteration). The re-planning strategies are methods to construct new plans with mechanisms whereby some plans properties are modified. A typical strategy in this calibration study is *ReRoute*. When this strategy performs, a complete new route is calculated for the passenger. The new plans are added to the agents' choice sets. Plans that achieved a poor score according to the utility-based approach are removed from choice sets.
- Choice: All other agents select between already known plans. Configurable selection strategies define the choice criteria. Typically the *ChangeExpBeta* strategy is used in standard simulations in order to approximate the simulation to a logit model.
- Go to: synthetic reality (above).

The iterative process is continued until some appropriate convergence criterion is reached. The loop describes day-to-day learning process.

4. Greater Berlin scenario

4.1. Demand preparation

The travel demand is modeled with the enumeration of each simulated traveler in the scenario. It includes the spatial and temporal description of their particular daily activities, like being home, work, shopping, etc. In this activity-based demand, the need of transport is derived from the required trips to connect travelers from one activity location to the next activity location.

The information about public transport demand in the Berlin and Brandenburg area was granted by BVG (Berlin public transport company). It corresponds to the demand that BVG is using in its own planning model. Neumann et al. (2014) describe how the demand was transformed from a macroscopic representation, corresponding to a PTV Vision (PTV AG, 2015) model, into a microscopic activity-based model description. On the macroscopic side, the (VIS)EVA sub-model of the PTV Vision suite was used for combined destination and mode choice. It was calibrated based on a large number of input data, most importantly a comprehensive trip diary survey of approximately 1% of the Berlin population (Ahrens et al., 2009), a specific public transit transport survey (VBB-Verkehrsverbund Berlin Brandenburg, 2008), and a pre-existing model of the metropolitan area (Berlin Senate Department for Urban Development and the Environment, and Brandenburg State Ministry for Infrastructure and Regional Planning, 2009). The microscopic demand, i.e. so-called initial plans for MATSim, was then generated by first generating a synthetic population based on the population categories of the macroscopic model, next assigning activity chains from the trip diary survey to each synthetic person based on socio-demographic attributes and home location, and finally drawing activity locations based on the OD matrices of the macroscopic model. Finally, the microscopic model carried out a relaxation process where route, time and mode choice dimensions were optimized so that results conform to certain macroscopic properties such as travel time distributions.

The plans of 598,891 persons who use all transport modes were adopted for the purposes of this work. In order to set the research focus on the transit calibration, the following additional preparatory steps were realized:

- Routing: as the original passenger's survey did not include data about selected routes, the routes between activity locations were calculated before the simulation. Basic route diversity was obtained by calculating 3 plans per agents with three travel priorities: strong walk penalty, strong transfer penalty and moderate values of them.
- Filtering: All persons who did not include the public transport mode at all in their plans where discarded. Some persons who intended to use public transport were also discarded, namely those whom the transit router calculated a direct walk to their destinations instead of a transit connection. 231,369 persons remained at the end.

Removing most of the car plans means that road congestion, and in consequence bus delays from road congestion, will be under-represented in the present study. This can be defended for the present study on the grounds that the Berlin public transit schedule already takes road congestion into account. The schedule-based transit simulation used in the present study takes this into account, by waiting at each stop for the scheduled departure time.

4.2. Counts

The occupancy counts refer here always to the observed number of passengers in transit vehicle after the stop, i.e. when the doors are finally closed for departure. Daily occupancy counts for stops were also granted by BVG. The occupancy measurements are defined on a per day basis. It means that all counts from simulation are not distributed in 24 time bins, but in one single time bin for a whole day.

A common situation in state estimation investigations is the availability of real data with different granularity grades in temporal and spatial dimensions (Nantes et al., 2015). For this work, these preparation steps were necessary:

- Filtering: From the original 7125 stop counts, only 2723 were selected for the calibration purposes. The big amount of discarded counts is explained by the fact that the data of aggregated stop counts and schedule transit stops were not directly linked because they originated from different projects. Some procedures were tried in order to relate each count with a stop: geographic concordance through coordinates, matching through resembling stops names and transit route pathways comparisons. Those procedures could validate only 2723 stops with a high degree of certainty.
- Stop zone conversion: The count data were not related to stops, but to stop zones. A stop zone describes a set of near stops that usually have the same name but each stop can be used by different transit lines or routes in different directions. Observed stop zone-based counts were not fragmented to assign occupancy values to each component stop because there was not any reference at all to do it. Instead of it, a module was developed to do stop zone-based analysis by aggregating the simulated particular stops occupancies into their respective zone. In fact, this demonstrates an advantage of the selected calibration approach: Any measurement that can be aggregated from the real world data can also be aggregated in the simulation, and thus any aggregation method can be used for calibration.

4.3. Transit schedule

The transit schedule contains the transport scenario supply data. The complete Berlin public transport network with 329 transit lines was used in the simulation.

4.4. Initial transit simulation

This section shows a preliminary standard simulation of the scenario data as starting point. First, transit routes for all agents were generated. Agents got their transit connections calculated with default parameter values of the transit router. The MUTTT and the MUTWT parameters include the opportunity cost of time. After all routes for passengers were calculated, a subsequent standard simulation run (i.e. a standard MATSim assignment) was executed without any calibration attempt. The evaluation of executed plans considers only the standard scoring function according to the transport behavioral model. Since that scoring function does not take crowding into account, the schedule-based transit assignment by itself should not change over the iterations. In the given model, small changes may happen from one iteration to the next, e.g. synthetic travelers not only using public transit may use a different car route and thus have a different travel time, and as a result public transit vehicles may leave a bit earlier or later because of other passenger boarding delays.

Fig. 1 shows the occupancy match analysis with scatter plots for the initial and final iterations of this initial simulation. In scatter plots, every dot represent both observed and simulated values for a stop at the given time bin. The observed passenger occupancy values are in the scale of X axis and the simulated occupancy values are in Y axis. Thus, a point in the diagonal would represent the state of exact match for both values. Expressed differently, for a perfect reproduction of counts in the simulation, all dots must be situated exactly over the diagonal with slope one. On the contrary, the more distant a dot from diagonal one is, the bigger is its observed-simulated values inconsistency.

One can notice that in both iterations, data points are dispersed outside the main diagonal, which is interpreted as under- and over-occupation. The Mean Relative Error (MRE; not shown as figure) for the simulated day and selected lines starts with

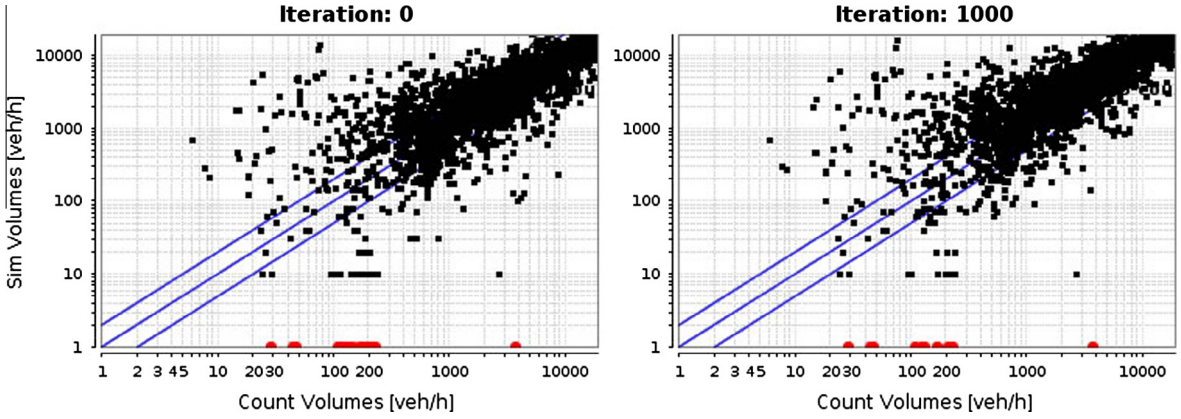


Fig. 1. Scatter plot for initial situation of Greater Berlin scenario: standard transit simulation without calibration.

89.6% and ends with 97.8% in iteration 1000. The reason why anything at all changes over the iterations lies in the fact that there is also a car traffic assignment, which changes over the iterations. This can, for example, cause synthetic travelers with mixed car/transit plans to obtain a different time structure because of changing car mode travel times. Overall, one can see that the initial simulation has room for improvement regarding the passenger occupancy concordance.

5. A calibration solution approach: Cadyts

5.1. Cadyts configuration

For transport simulations, but with the clear potential to be more general, the demand estimation problem was addressed by G. Flötteröd. He implemented his methodological approach into the open source software Cadyts. Intuitively, the approach uses the freedom that is left when individual decisions are modeled as random draws from a discrete choice model: Decisions that are congruent with the observations become preferred over those that are not.

Cadyts theoretical principles can be found in (Flötteröd et al., 2011a). In this section, only the calibration steps are briefly introduced:

1. Initialization. The calibration process starts by registering observed counts at stops. Each entry has this data structure:

- Id: defines the unique identifier for the count station.
- Start time: defines the inclusive initial time for the count time bin.
- End time: defines the exclusive final time for the count time bin.
- Value: defines the number of observed mobile agents during the time bin.
- Minimal standard deviation (`minStdDev`): defines the smallest allowed standard deviation for the observed counts. After an initial experimentation, it was set to 4 for tests described in this work; see below for more information.

2. Choice. After a given simulation iteration, the calibrator calculates the utility correction per agent that influences the plan choice to match the observed counts in a more efficient way. The choice is based on the calculated utility correction and reported to the calibrator. For the purpose of this work, a special implementation makes Cadyts act as a weighted accumulated scoring function.

3. Update. The calibrator is informed about the network loading conditions after the simulation iteration.

The Cadyts posterior choice model is outlined in Section 3.1 of Flötteröd et al. (2011b). Section 3.2 there presents the formulation of its distribution embedded with the MATSim demand simulation multinomial logit model. It assumes moderate congestion with independently normal distributed traffic counts. The core equation for the purposes here is:

$$P(i|y) \sim \exp\left(V(i) + \sum_{ak \in i} \frac{y_a(k) - q_a(k)}{\sigma_a^2(k)}\right) \tag{2}$$

where

y is a vector that collects all actual data.
$P(i|y)$ is the posterior plan choice distribution given y.
$V(i)$ is the standard score of a plan i as formulated in Eq. (1).
$y_a(k)$ is the actual observed count at location a during time k.
$q_a(k)$ is the simulated count at location a during simulated time k.
$\sigma_a^2(k)$ is the inverse weight of the measurement.

The inverse weight $\sigma_a^2(k)$ should optimally come from specific knowledge about each sensor, but in its absence, it is calculated as: $\sigma_a^2(k) = \max(varianceScale * y_a(k), minStdDev^2)$, i.e. assumed to be proportional to the measured value in order to be consistent with an assumption of Poisson distributed measurements. For the present paper, variance Scale is left at its default value of 1.0, meaning that all measurements below $y_a = 16$ are under-weighted because $minStdDev = 4$.

5.2. Initial brute force calibration with fixed route choice

The first calibration attempt over the scenario is presented in this section. It was realized with the brute force setting, which enforces the reproduction of counts with best effort, usually overriding the effect of other evolutionary mechanisms like the MATSim scoring function. The brute force is not a plausible method in a behavioral context, but it is used only as preliminary test to prove the capacity of calibration settings over the synthetic demand in its initial state.

Similar to (Moyo and Nagel, 2012), a fixed set of three transit connection alternatives per passenger was pre-calculated according to three diverse travel criteria: least number of interchanges, least amount of walking, and a balance of them. Cadyts then selects between these three routes. For this first calibration, Fig. 2 shows the comparison of counts adequacy between initial iteration 0 and final iteration 1000 in scatter plots.

One can see that the brute force calibration ends up with an acceptable match of counts, with a large number of agents and day-based counts. The brute force setting pushes the simulation into the reproduction of occupancy volumes by selecting the plans most consistent with the observations. The MRE starts with value 128.5% and it is reduced to 15.3% at iteration 1000.

However, while the approach works well, the brute force option and the scoring function switching-off are incongruous from a behavioral modeling perspective. Moreover, the method depends on the fixed set of pre-calculated passenger routes. Without plans mutation, Cadyts can only shift between existing plans and this would not allow the calibration procedure to guide the search into the reproduction of real measurements.

5.3. Transit route diversity research

It is assumed that the calibration core procedure leaves the generation of choice alternatives to the simulation counterpart. Cadyts only influences the reproduction of counts by evaluating and preferring some alternatives that are presented to it. At the end, a favorable calibration output depends to a large extent on the generation of sufficient choice diversity. A reduced number of routes, or a choice set that do not correspond to realistic passengers routes, will affect negatively the expected count match.

Two ways have been so far considered to apply Cadyts to agent-based public transit assignment. In a first approach (Moyo and Nagel, 2012; also described above), multiple transit connection alternatives per passenger were pre-calculated. A second approach is presented here, where new plans are generated as part of the choice set modification step (Step 4). With the goal of discarding the pre-calculation of the route set and generate discretionary connections instead, a special routing module was implemented in MATSim. The "randomized transit router" takes MUTWT, MUTTW, MUTTT, MUTDT and ULS default values and generates new random travel cost coefficients. Every time that each parameter gets a new random value, a new different connection can be generated. In this way, very diverse passenger travel priorities are simulated.

The generation of route diversity from random values is illustrated with an example of a simulated person who needs to travel from the TU Transport Systems Planning and Transport Telematics Institute to the transit hub *Alexanderplatz* in Berlin. First, the transit router calculates an initial route with default parameter values (the opportunity cost of time is included). With random re-routing as re-planning strategy, after 10 more iterations, 11 combinations of random values are generated. They are shown in Table 2.

At each iteration, the respective value combination is given to the transit router to calculate a new additional connection for the passenger. These 11 routes are graphically represented in Fig. 3.

The origin is marked with letter "0" and the destination with "D". For 11 queries, the randomized transit router calculated 8 different routes with diversity in number of transfers and walk distances. Actually, the first route in light green with number 0 is a direct walk to the destination, which means that the initial values are not good enough to find a connection for this example. The same color is employed for the starting and ending walking distances in the other routes. Four iterations calculated the same route marked with a red line. It can be explained with the fact that that connection corresponds to the use of a near bus line that links to the high frequency and fast suburban metro lines to *Alexanderplatz*.

The approach, as described here, will use the same randomization for each traveler. That is, every time the transit router is called, it operates with a different set of relative weights for the cost components, but those weights are always drawn from the same statistical distribution. In the co-evolutionary MATSim approach, it is the task of the scoring function to finally select the option that is most appropriate for each specific traveler. For example, with respect to Fig. 3, a person with a high dislike of walk would avoid route 0, a person with a high dislike of transfers would prefer route 6, etc. Nagel et al. (2014) describe how such an approach in the context of car and truck route choice can be put into practice; also the conclusion of the present paper alludes to methods of how to achieve this.

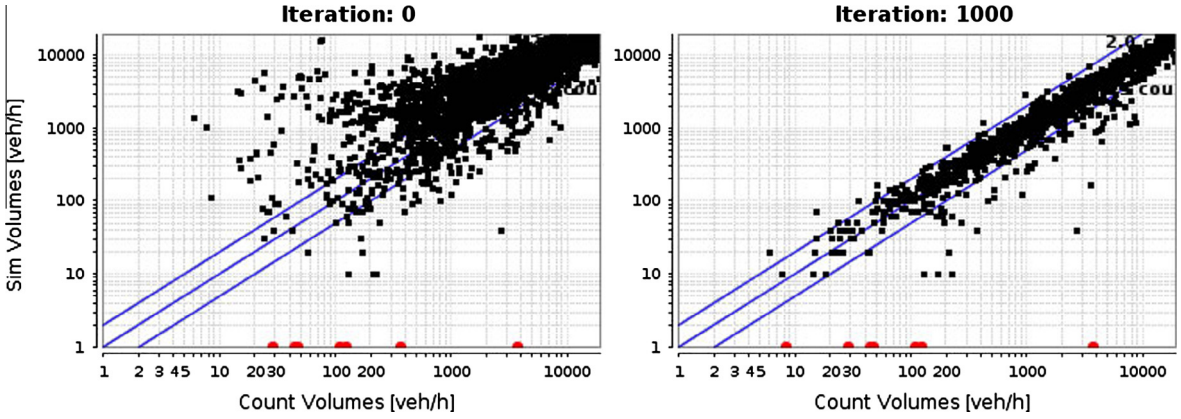


Fig. 2. Brute force calibration of Greater Berlin scenario with fixed route choice.

Table 2
Travel parameter random value generation example.

Random values combination	MUTTW	ULS	MUTWT	MUTTT
Default	−12.0/3600	−1.0	−12.0/3600	−12.0/3600
0	−2.6/3600	−1.9	4.5/3600	−18.3/3600
1	−13.0/3600	−1.3	−19.4/3600	−9.0/3600
2	−59.9/3600	−3.2	−9.6/3600	−43.8/3600
3	−39.1/3600	−2.6	−9.0/3600	−59.6/3600
4	−26.0/3600	−4.5	−29.2/3600	−34.3/3600
5	−20.7/3600	−3.7	−10.1/3600	−27.9/3600
6	−7.6/3600	−0.6	−0.3/3600	−2.6/3600
7	−46.8/3600	0.0	−29.7/3600	−18.5/3600
8	−33.8/3600	−1.9	−19.9/3600	−53.2/3600
9	−26.5/3600	−1.9	−14.9/3600	−16.0/3600
10	−13.4/3600	−3.7	−5.1/3600	−50.7/3600

5.4. Cadyts calibration as scoring function

Cadyts calculates utility corrections for plans in order to guide the re-planning and choice process in the direction of the count match. The new implementation of MATSim-Cadyts integration presented here consists in the integration of the calibration linear plan effect with the other simulation scoring components (performing and traveling). Thus, the count match is also included as part of the plan evaluation. More formally, the Cadyts core function of posterior choice distribution (Eq. (2)) is added to the utility of plan $V(i)$ (Eq. (1)) during the scoring process as a new weighted term:

$$V(i) = \sum_{act \in i} \beta_{perf} \cdot t^*_{act} \cdot \ln\, t_{perf,act} + \sum_{leg \in i} V_{tr,leg} + \left(w \sum_{ak \in i} \frac{y_a(k) - q_a(k)}{\sigma^2_a(k)} \right) \tag{3}$$

where

w is the weight of Cadyts correction inside the accumulated scoring function.

Having Cadyts as part of the scoring function leads to these advantages:

- Brute force is technically abandoned which returns the calibration standpoint to a behavioral model.
- Both plans performance and their count match contribution can be evaluated together with compound utility formulation.
- An important point is related to the MATSim evolutionary algorithm: If an agent ends up with more plans than it is allowed to have, plans are removed from that agent until it is back to the permitted number. The rule according to which these plans are selected obviously has a strong influence of the composition of the set of plans that an agent has to its immediate disposal. When the Cadyts offset is added to the score, then that combined score forms the basis of the removal decision. In consequence, good plans from the calibration perspective can then persist over the iterations, which influences positively on the calibration feature. In contrast, a scoring model based solely on travel disutility and activity performance evaluation jeopardizes the existence of plans that are plausible for the count concordance. That is, plans that are dismissed when not including the Cadyts offset into the score are often the ones that are considered the worst from the standard behavioral scoring context, overlooking their contribution to count reproduction.

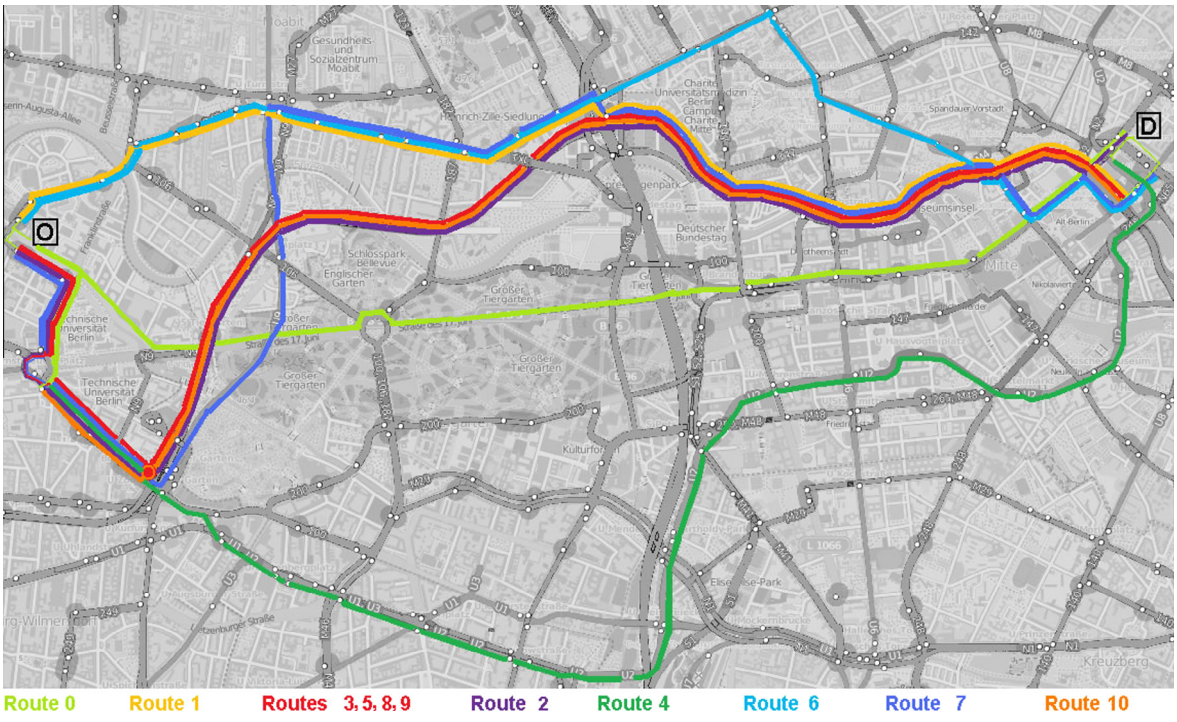


Fig. 3. *Randomized transit router* example: 11 route queries from TU Transport Systems Planning and Transport Telematics Institute to transit hub *Alexanderplatz* in Berlin. *Source:* www.openmap.lt.

- The adjustable value *w* represents the relative importance of the utility correction on the general model. Weighting the calibration has advantages in comparison with the explicit brute force setting. With this configurable calibration strength value, other plan evaluation components are not completely overridden. The effect is comparable to the variance scale parameter in Cadyts, but has the advantage that it is not overridden by the *minStdDev* parameter which effectively neutralizes any effect of a small *varianceScale*.

5.5. Coupling route diversity and Cadyts scoring function

Achieving route diversity through random routes generation might seem inadequate from the classical assignment models perspective. Certainly it would be impractical if it were implemented as a stand-alone module for route choice model. However, its implementation is justified because random paths are created on the base of proved standard travel values as initial seed. But most important, if the search of random candidate solutions is combined with a selection mechanism (like the scoring function) where new alternatives for each agent are evaluated and the worst are discarded, this coupling constitutes a composite co-evolutionary algorithm that directs the choice distribution to a count match convergence. The integration of both approaches is graphically represented in Fig. 4 and outlined next.

- Initialization: Usual scenario data are loaded, including the revealed occupancy counts. Some of the configurable parameters in this step are: calibration weight, maximal number of plans per agent, execution probability of each strategy, and use of stop zone conversion for analysis.
- Initial routing: The router generates the first alternative route plan and it is selected.
- Execution: The selected plan is executed. The simulation includes vehicular traffic flow simulation.
- Scoring: The executed plan is evaluated according to the accumulated scoring function (Eq. (3)).
- Re-planning: New routes might be calculated (according to the re-routing strategy probability), and worst plans from behavioral and count convergence perspectives are discarded.
- Selection: A plan is selected and scored if it was never executed before. If all plans of an agent are scored, the *ChangeExpBeta* strategy selects the plan (generating a logit distribution).
- Iteration: The process goes back to "execution".
- Analysis: It includes the MRE calculation and generation of count juxtaposition graphs.

For the implementation of the approach on the Greater Berlin scenario, further adaptations and settings are reported:

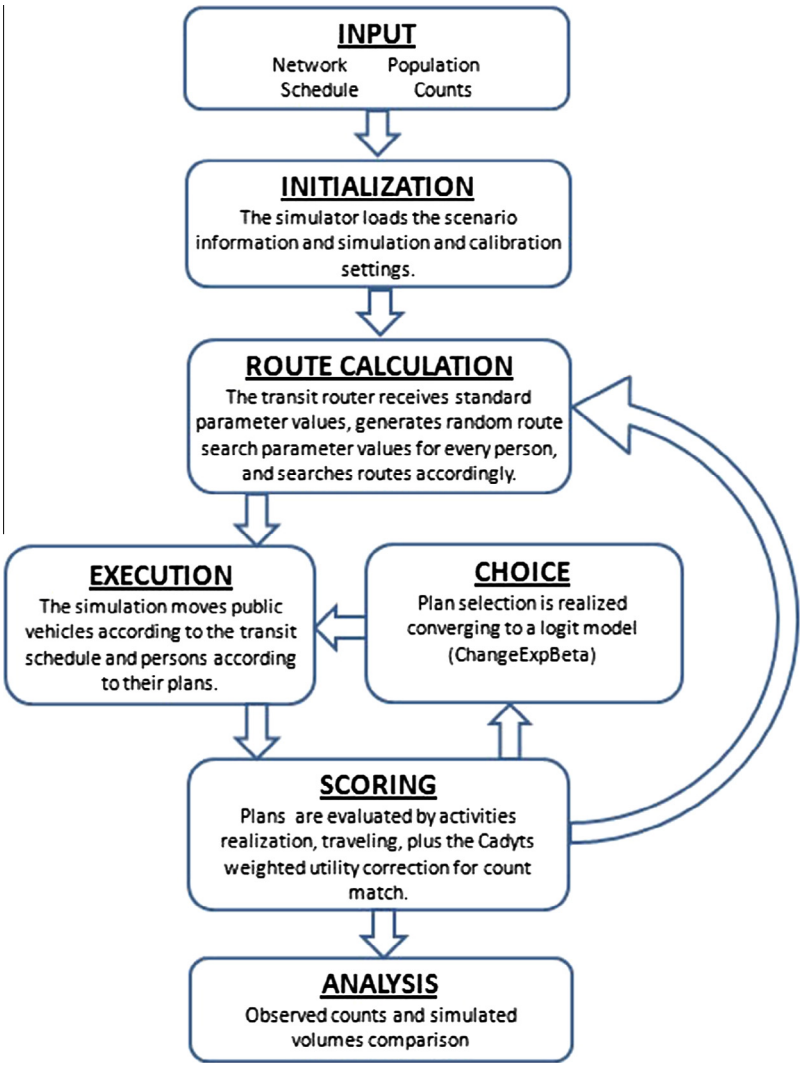


Fig. 4. Diagram of randomized routing plus Cadyts inside the scoring function.

- Cadyts calibrator instance called "Analytical Calibrator" was used in this experiment to allow the creation of a calibrator object with the configured 86,400 s sized time bins per station.
- Calibrated hours: the complete day from 0 to 24 (in concordance to the 24-h counts).
- Maximal number of plans per agent: 5.
- Simulation and calibration settings: Cadyts utility correction is embedded in the scoring function. The scoring performs as usual for each plan that is selected at a given iteration. The simulation re-planning module is configured to distribute its execution probabilities like this: randomized re-routing with 10% of probability until iteration 400 and *ChangeExpBeta* as plan selection strategy with 90% of probability. *ChangeExpBeta* keeps performing until iteration 1000, along with the scoring approach that includes Cadyts correction utility.

6. Results

In order to evaluate how Cadyts enforces the reproduction of counts, a number of parametric calibration runs is realized over the scenario. The runs are realized with incrementing calibration weights (w in Eq. (3)), namely 0, 1, 10, 100, and 1000. Fig. 5 shows the analysis in scatter plots for each run. The first plot corresponds to the initial iteration, which is the same for all weights. Then, the output plot (iteration 1000) for each different weight is depicted.

One can see that for low calibration weights (like 0, 1, and 10) hardly any improvement is achieved regarding count match. On the contrary, it is noticeable that a very strong value like 1000 corresponds almost to the brute force calibration.

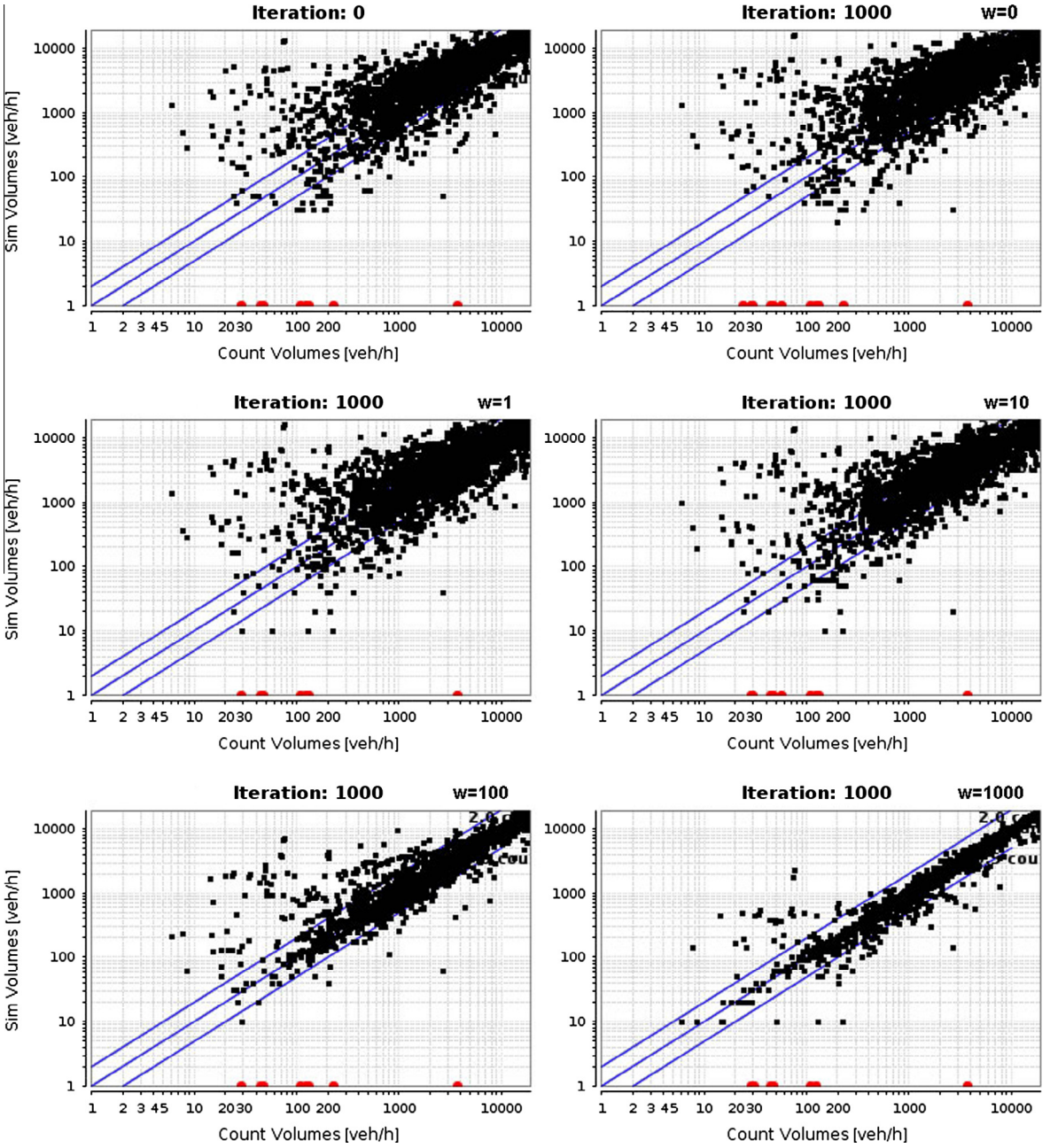


Fig. 5. Calibration with randomized parameters route search and Cadyts as part of the scoring function with increasing weight.

Another simulation exercise is described next, whereby the synthetic demand is duplicated, that is, each agent is cloned. Moreover, instead of 5 plans per agent, the choice set size was increased to 10. With it, more diverse routes are available for passengers and also for the calibrator. Fig. 6 shows the initial and final plots for the calibration with cloned agents.

Another analysis consists on tracking the MRE value for observed and simulated count divergence along iterations. Table 3 shows the initial and final MRE value for all runs presented in this work.

One can notice that higher Cadyts weights achieve lower final MRE values. In the same way, Fig. 7 compares graphically evolution of the MRE values along iterations of all calibration tests described before. Values after iteration 700 do barely change and are thus omitted to improve the resolution of the plot.

The calibration with brute force with fixed choice set starts with a high MRE value (128.5%), in comparison with the value (100.0%) of all runs made with randomized routing. This can be explained by the fact that the fixed choice set starts always with the first plan selected, that corresponds to the pure high transfer resistance. However, the fixed plans and the plans created with random routing tend to stabilize and come relative to close values (42% and 15% respectively) around iteration 600.

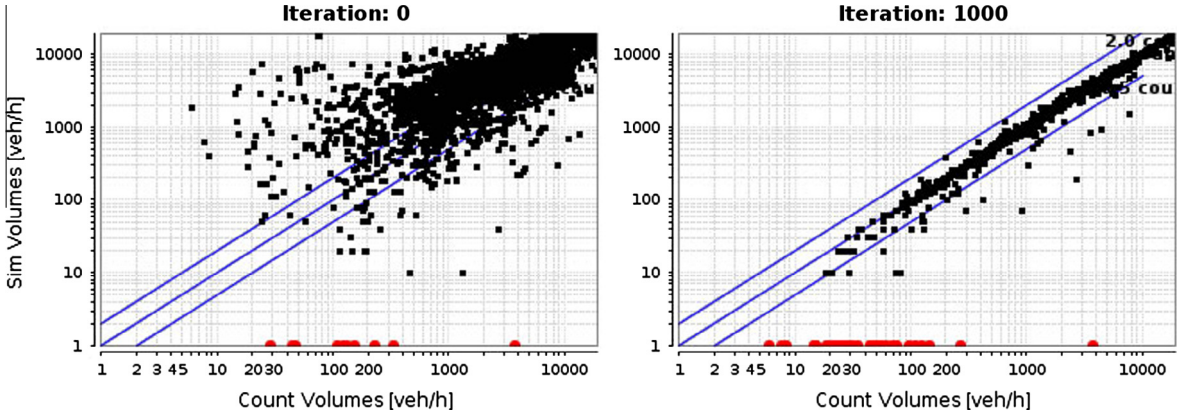


Fig. 6. Calibration of Greater Berlin scenario with duplicated demand and Cadyts weight 1000.

Table 3
Comparison of initial and final MRE values for calibration runs.

Test	Initial MRE	Final MRE
Simulation initial situation (without calibration)	89.6	97.8
Brute Force calibration with fixed choice set	128.5	15.1
Random routing and Cadyts weight 0	100.0	107.2
Random routing and Cadyts weight 1	100.0	104.4
Random routing and Cadyts weight 10	100.0	89.8
Random routing and Cadyts weight 100	100.0	41.5
Random routing and Cadyts weight 1000	100.0	15.7
Random routing and Cadyts weight 1000 with agent cloning	151.3	5.0

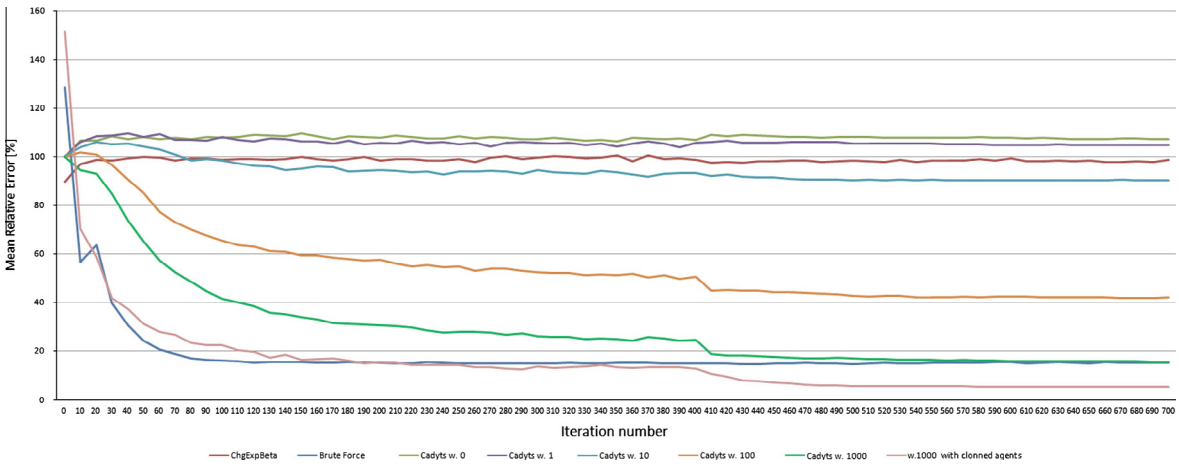


Fig. 7. Evolution of MRE values along iterations with normal simulation, brute force calibration and Cadyts as scoring function with different weights and duplicated demand.

The analysis also shows that low Cadyts weight values have barely a visible effect. In contrast, one may see the evident improvement for values 100 and 1000 just after the first iterations. In both strong calibrations, a sudden error reduction that happens just after iteration 400 is noteworthy. It corresponds to the stop of plan innovation (with randomized router) and the start of full calibration. The strongest weight value (1000) deserves special attention because it reaches the same MRE value of brute force calibration from iteration 600 on.

The calibration of duplicated demand starts with the worst count reproduction, but the effect of the strongest Cadyts weight value and the larger number of agents produces at the end the best output.

The results obtained with the brute force option and the higher weights require special attention. Although similar error reduction pattern are achieved with them, the technical differences are:

- The brute force option is only a useful configuration that can help to identify inconsistent measurements that cannot be reproduced, even when this option is used. In this work it was used as preliminary exercise over the scenario. However, it does not represent a meaningful option from a behavioral or model calibration point of view.
- Other internal calibration settings like the minimal standard deviation or variance scale try to weight the measurements accordingly too. Unfortunately, these parameters were not enough to achieve a strong error reduction outside the brute force.
- The inclusion of the calibration utility correction presented here reduced the error almost like the brute force option. But its notorious advantages are: it is applied inside a behavioral context and it is executed along an automatic route creation mechanism.

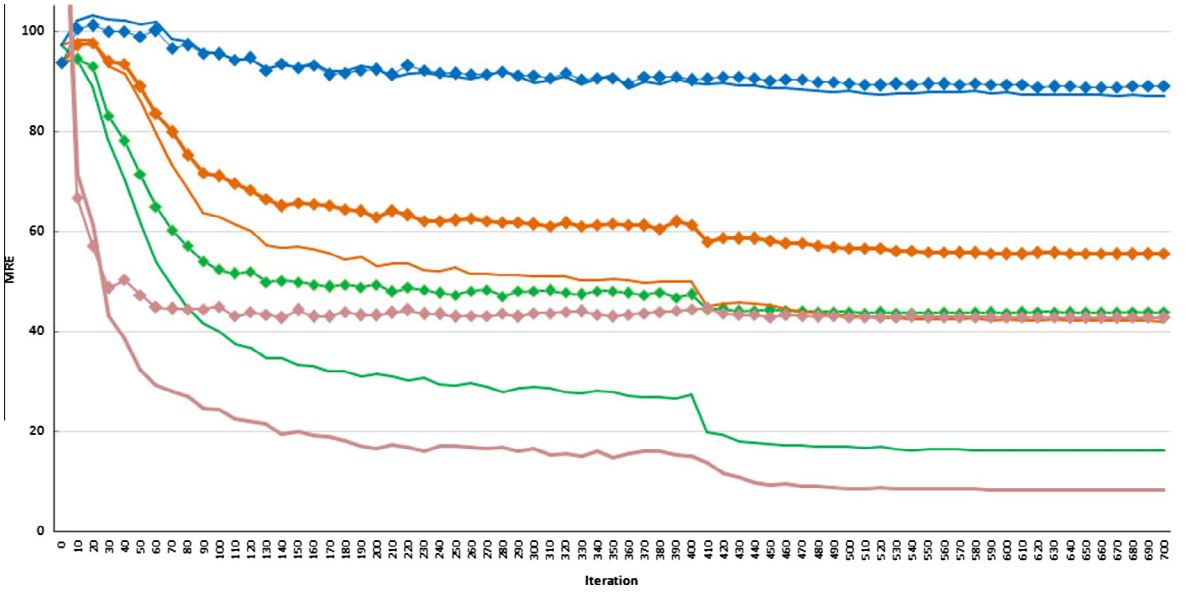
7. Public transit passenger occupancy feedback over non-calibrated transit stops (cross validation)

One essential question in assessing the count match adjustment is its impact over the whole system. Modifications in the calibrated stations lead to changed passenger occupancy patterns in the complete network load, which then also affect those stations without observed measurements.

Moreover, it is especially relevant to make sure that the seed joint distribution of trips across multiple dimensions (mode, route, time, location) remains well-preserved along the complete adjustment process.

This section reports an exercise whose goal was to ensure that the proposed calibration method does not generate any other bias that may produce unforeseen collateral effects. First, the available occupancy counts were randomly divided into two groups with the same number of counts:

1. Adjustment counts: The first group was used as basis for utility function adjustment, in the same way as it is described in last section.
2. Control counts: The second group was not introduced into the calibrator and it was used only as a control group for the post-adjustment analysis of non-calibrated stations.



Adjustment counts			
Weight	Line	Initial MRE	Final MRE
10		97.3 %	87.0 %
100		97.3 %	42.0 %
1000		97.3 %	16.3 %
1000 double demand		145.3 %	8.4 %

Control counts			
Weight	Line	Initial MRE	Final MRE
10		93.7 %	89.1 %
100		93.7 %	55.5 %
1000		93.7 %	43.8 %
1000 double demand		142.9 %	42.7 %

Fig. 8. Evolution of MRE values along iterations with two count groups: adjustment counts and control counts with different weights and duplicated demand.

In concordance with the experiments executed over the scenario in last section, the calibration with the adjustment counts was applied with weights: 10, 100, and 1000. Similarly, the weight 1000 was applied also on the duplicated demand. After that, a posterior occupancy analysis considered only the control counts. Fig. 8 shows the evolution of MRE reduction along iterations, for both adjustment and control counts.

One can see that the calibration with the low weight value 10 leaves the error levels for both groups at approximately the same level. In this case it is confirmed that the calibration does not introduce any bias to the system. However, this low weight does not produce a meaningful match progress over the control group either. It is only with higher weight values that the calibration also reduces to same extent the error over the control group as well. It is notable, however, that this positive side-effect occurs mainly before iteration 150 in control counts. The sudden error reduction at iteration 400 that was described in the previous section is also present here in control counts with weights 100 and 1000. Overall, this result confirms that the inserted calibration process makes an improvement on the model itself.

8. Conclusions and future work

The calibration of a large scenario was carried out, preparing the necessary input data and adapting the integration code between transit simulation and calibration. This is possible thanks to the adaptability of Cadyts (proved with its many different implementations) and the robustness of the transit simulation in MATSim.

The results demonstrate that the calibration approach is able to work with very large scale, real world scenarios, and that it is able to deal with the inter-temporal aspects implied by the available counts.

The contribution of the proposed approach is that it puts aside the mainstream passenger flow estimation and goes into the microscopic behavioral level to give detailed information of each passenger track record. This opens the possibility of completely disaggregated or collective demand estimation analysis.

Demand estimation on individual level could have practical applications. More reliable travel demand information obtained from a behavioral model calibration can be of benefit for travel agencies that are already able to collect detailed transit usage information. For example, it is very useful to study the interaction between schedule stability measures and travel demand involving observed stations (Wang et al., 2015). For those cases the demand should be more realistic in terms of observed travel usage. The proposed approach is very suitable for use in those cases.

In this sense, the next challenge will be how to make these findings useful for prediction. The approach for this will be to extract behavioral parameters, possibly per individual, which would explain behaviorally the choices that are most consistent with the measurements. An option is the socioeconomic stratification according to internal homogeneity, which has been studied in choice models for long time. Another idea is to parameterize the adjustments rather than calculate them rigorously on individual level. That could be managed with the calculation of average corrections according to population stratification (by income, mode, age, activity pattern, etc.). It could be useful for policy studies where that transit users might show a different travel behavior in comparison to the base year.

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